Placeholder University

Placeholder Doctoral School
Information Sciences & Technologies

Doctoral Thesis

Presented by:
Fourth Student

A Placeholder Title for a Placeholder Thesis

Defended publicly on **September 30th, 2026** before the jury composed of:

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Table of Contents

Acknowledgements

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This chapter carries no number, and does not appear in the mini table of contents of the chapters that follow. It can still be referred to like any other one: `\nameref{phd_acknowledgements}` gives Acknowledgements.

Chapter I

Introduction

A PhD manuscript uses exactly the same commands as an HDR one: only the `phd` option of the document class differs.

This chapter introduces the placeholder problem addressed in this thesis, and outlines the rest of the document.

I.1 Context

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua [Ipsum *et al.*, 1998]. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat [Tempor *et al.*, 2009].

The rest of this document uses ML vocabulary, and in particular the Convolutional Neural Network (CNN). Note that CNNs is a separate entry of `acronyms.tex`, whose only purpose is to provide a plural tooltip: its full name is Convolutional Neural Networks, and it does not appear in the list of acronyms. Recall that State Of The Art (SOTA) results in this placeholder field are due to [Amet *et al.*, 2017].

I.2 Problem Statement

Problem 1 Placeholder problem

Given a set of placeholder observations $\{\boldsymbol{x}_i\}_{i=1}^N$ and their placeholder targets $\{\boldsymbol{y}_i\}_{i=1}^N$, find a placeholder model f_θ that generalizes to unseen placeholder observations.

Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Problem 1 is addressed in Chapter III, after the review of Chapter II.

I.3 Outline

Chapter II reviews the placeholder literature. Chapter III presents our contribution, evaluated in Section III.2. Chapter IV concludes, and Appendix A gathers the proofs. The document opens on the Acknowledgements and closes on the Bibliography and the list of Acronyms: these three chapters are unnumbered, and are therefore recalled by their title rather than by a number.

Chapter II

State of the Art

This chapter reviews the placeholder literature, and positions the contribution of Chapter III with respect to it.

II.1 Early Approaches

Lorem ipsum dolor sit amet, consectetur adipiscing elit [Ipsum *et al.*, 1998]. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua, as summarized in the textbook of [Tempor *et al.*, 2009].

Definition 1 Placeholder representation

A *placeholder representation* of an observation $\boldsymbol{x}$ is any vector $\boldsymbol{z} = g(\boldsymbol{x})$ such that $\boldsymbol{z}$ carries no more information than $\boldsymbol{x}$.

II.2 Modern Approaches

Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat [Amet *et al.*, 2017]. CNNs are now trained on a single GPU, which was not the case for the earlier approaches of Section II.1. Figure 1 summarizes the pipeline they all share.

Figure 1 An overview of the placeholder pipeline shared by modern approaches. The same figure is reused in Chapter III, which is perfectly fine as long as the labels differ.

II.3 Summary

Table 1 compares the approaches discussed above. None of them addresses Problem 1 directly, which motivates the contribution of Chapter III.

Approach	**Placeholder**	**Scalable**	**Interpretable**
[Ipsum *et al.*, 1998]	✓	✗	✓
[Tempor *et al.*, 2009]	✓	✗	✗
[Amet *et al.*, 2017]	✓	✓	✗
Ours	✓	✓	✓

Table 1 A comparison of placeholder approaches. Checkmarks are placeholders too.

Chapter III

A Placeholder Contribution

This chapter presents the contribution of this thesis, and is the one that demonstrates most of the environments provided by the template.

It answers Problem 1, stated in Chapter I.

III.1 Method

We look for a placeholder model f_θ minimizing the objective of Equation 1, where $\lambda > 0$ controls the strength of the placeholder regularizer.

$$\mathcal{L}(\theta) = \underbrace{\frac{1}{N}\sum_{i=1}^{N} \|f_\theta(\boldsymbol{x}_i) - \boldsymbol{y}_i\|_2^2}_{\text{placeholder fidelity}} + \lambda \underbrace{\|\theta\|_2^2}_{\text{placeholder regularizer}} \tag{1}$$

Proposition 1 Existence of a placeholder minimizer

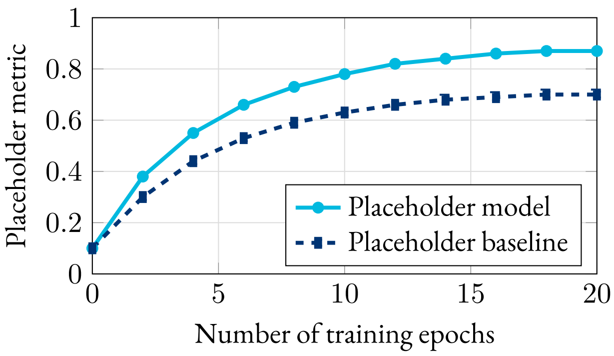
For every $\lambda > 0$, the objective of Equation 1 admits at least one minimizer $\theta^\star$.

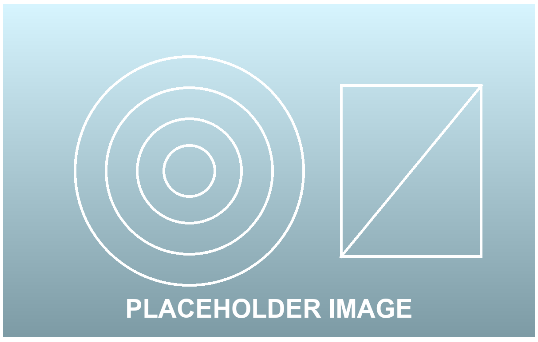
The proof of Proposition 1 is deferred to Appendix A, so as not to interrupt the exposition. An unnumbered equation is obtained with the starred variant, which is convenient for intermediate steps:

$$\nabla_\theta \mathcal{L}(\theta) = \frac{2}{N}\sum_{i=1}^{N} (f_\theta(\boldsymbol{x}_i) - \boldsymbol{y}_i)\, \nabla_\theta f_\theta(\boldsymbol{x}_i) + 2\lambda\theta$$

III.2 Results

We evaluate the method of Section III.1 on placeholder datasets. Figure 2 reports the learning curves, and Table 2 the final scores.





(b) A qualitative placeholder example.

(a) Placeholder metric along training, for our model and the baseline of [Amet *et al.*, 2017].

Figure 2 Placeholder results. (a) reports the metric along training, while (b) shows a qualitative example.

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Dataset	[**Amet *et al.*, 2017**]	Ours
Placeholder-A	0.70 ± 0.02	$\mathbf{0.87 \pm 0.01}$
Placeholder-B	0.61 ± 0.03	$\mathbf{0.79 \pm 0.02}$
Placeholder-C	0.55 ± 0.04	$\mathbf{0.68 \pm 0.03}$

Table 2 Final placeholder scores, averaged over five placeholder runs. Best results in bold.

III.3 Discussion

Limitations

An unnumbered box, obtained with `boxenv*`, is handy for remarks that do not need to be cross-referenced. The placeholder method has placeholder limitations, which are discussed in Chapter IV.

Chapter IV

Conclusion & Perspectives

A short closing chapter, summarizing the contribution of Chapter III and listing the directions it opens.

IV.1 Summary

This thesis addressed Problem 1, introduced in Chapter I. After the review of Chapter II, Chapter III proposed a placeholder method, whose objective is given in Equation 1 and whose results are reported in Table 2.

IV.2 Perspectives

- Extending the placeholder method to placeholder data of higher dimension.
- Reducing the GPU cost of placeholder training, so that the approach scales beyond placeholder datasets.
- Studying the interpretability of the placeholder representations of Definition 1.

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Appendix A

Proofs

Deferring proofs to an appendix keeps the main text readable.

The `proof` environment takes the label of the statement it proves, and displays a clickable reference to it in its title.

A.1 Existence of a Placeholder Minimizer

Proof of Proposition 1

The objective of Equation 1 is continuous in θ, and coercive as soon as $\lambda > 0$, since $\|\theta\|_2^2 \to \infty$ when $\|\theta\|_2 \to \infty$. A continuous coercive function on $\mathbb{R}^d$ attains its infimum, hence the existence of a minimizer $\theta^\star$. $\square$

A.2 A Proof Without a Statement

The mandatory argument of `proof` can also be left empty, in which case the box is simply titled *"Proof"*.

Proof

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Bibliography

References I Contributed To

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10

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Second Coauthor, Firstname Lastname (2024). **"Ut Enim Ad Minim Veniam: Scaling Placeholder Models on a Single GPU"**. *Transactions on Placeholder Systems (vol. 7, pp. 1–21).*

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— 6, 8, 10, 11

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[Ipsum *et al.*, 1998]
— 6, 8

Lorem Ipsum, Sit Dolor (1998). **"A Foundational Placeholder Study of Nothing in Particular"**. *Proceedings of the Placeholder Society (vol. 3, nb. 2, pp. 17–29).*

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Online Resources Cited in this Document

[GitHub]
August 19^{th}, 2026

Bastien Pasdeloup. **"A LaTeX Template for HDR and PhD Manuscripts"**. *GitHub.*

[Placeholder]
August 19^{th}, 2026

"A Placeholder Online Resource". *Placeholder Organization.*

Acronyms

AI	**A**rtificial **I**ntelligence
BSc	**B**achelor of **Sc**ience
CNN	**C**onvolutional **N**eural **N**etwork
GNU	**GNU**'s **N**ot **U**nix
GPL	**G**eneral **P**ublic **L**icense
GPU	**G**raphics **P**rocessing **U**nit
HDR	**H**abilitation à **D**iriger des **R**echerches
ML	**M**achine **L**earning
MSc	**M**aster of **Sc**ience
PDF	**P**ortable **D**ocument **F**ormat
PhD	**Ph**ilosophiæ **D**octor
SOTA	**S**tate **O**f **T**he **A**rt

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